Asking RMBS loan data questions in plain English: results

Utkarsh Malhotra · November 2026

A language model turns each question into a small query spec, and a compiler turns the spec into SQL over governed metric definitions, so the model never does the arithmetic. These results are from a working prototype on synthetic data: 50 Australian RMBS deals, 21,945 loans and 773,336 loan-months in 2 trustee formats.

What is proved

check	result
Cleaning: canonical table against simulated truth	0 mismatches over 773,336 loan-months
Cleaning: resubmitted duplicates removed	721 of 774,057 raw rows
Cleaning: clean-up calls coded as repayments, recovered	7 deals
Compiler: output against hand-written SQL	49 of 49 match, worst difference 1.6e-15
Compiler: queries that render to Oracle	49 of 49
Scale: slowest question at 27,840,096 loan-months	under 0.75 s with default threads

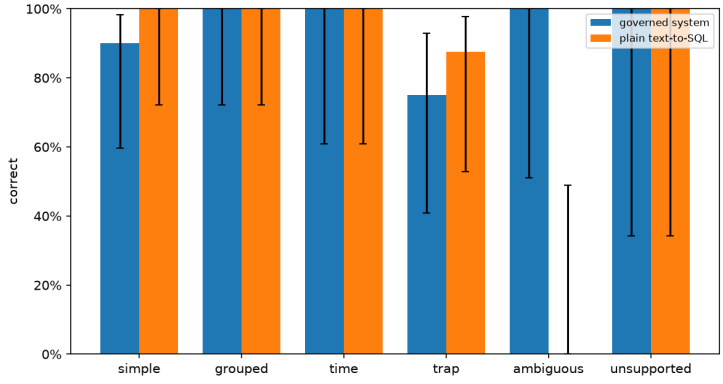
The traps in the questions, measured on this data

trap	question	correct	naive mistake gives
Arrears units	Q27	0.66%	0.59%
Calls as prepayments	Q29	21.6%	100.0%
Calls as prepayments	Q30	18.3%	49.0%
Exit rows as live loans	Q32	9,975	10,572
Unweighted average	Q33	51.96%	50.92%
Averaging CPR (dropped, gap too small)	Q2 2025 CPR	18.064%	18.063%

Governed system against plain text-to-SQL

Same 40 questions. The plain model's prompt held the same table definitions and written metric formulas. Neither system returned a wrong number: every miss on both sides was a refusal. The governed system asked on every ambiguous question, where plain SQL declined without saying why.

category	questions	governed system	plain text-to-SQL
simple	10	9 (90%)	10 (100%)
grouped	10	10 (100%)	10 (100%)
time	6	6 (100%)	6 (100%)
trap	8	6 (75%)	7 (88%)
ambiguous	4	4 (100%)	0 (0%)
unsupported	2	2 (100%)	2 (100%)
total	40	37 (92%)	35 (88%)



Synthetic data. Source: outputs/eval_summary.csv.

Asked "What are arrears in NSW?", the governed system replied:

```
Which arrears measure do you want for NSW?
Options: 30+ days past due; 90+ days past due
```